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Steering apparatus

Abstract

A steering apparatus is for turning steerable wheels of a vehicle. The steering apparatus includes a steering wheel rotatable about a steering axis to effect turning of the steerable vehicle wheels. A first shaft is connected to the steering wheel and rotatable about a first shaft axis. A first nut is threadably connected to and longitudinally movable along the first shaft. A second nut is connected to the first nut such that the first and second nuts are prevented from moving longitudinally relative to one another. A second shaft is rotatable about a second shaft axis. The second nut is threadably connected to and longitudinally movable along the second shaft. A motor is connected to the second shaft and controllable to apply a force to the steering wheel through the first and second shafts and the first and second nuts.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3703835	12/1971	O'Connor	74/441	F16H 25/2006
3716110	12/1972	Fonda	180/428	B62D 5/04
5174407	12/1991	Shimizu	180/444	B62D 1/166
2021/0309277	12/2020	Huber et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
1767432	12/2006	EP	B62D 1/181
2005067386	12/2004	JP	N/A
2009292331	12/2008	JP	N/A
2012045978	12/2011	JP	N/A

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to a steering apparatus for turning steerable wheels of a vehicle in response to rotation of a vehicle steering wheel.

BACKGROUND OF THE INVENTION

(2) Power steering assemblies are common in modern vehicles. Typically, one or more rigid shafts connect a vehicle steering wheel to a steering rack of the power steering assembly. The rigid shafts must be routed from the vehicle steering wheel to the steering rack. Routing the rigid shafts between the steering wheel and the steering rack is often difficult, as other vehicle components must not interfere with the shafts.

(3) Some known vehicle steering systems have eliminated the rigid shafts. Such systems are commonly referred to as “steer-by-wire” systems. In steer-by-wire systems, there is no mechanical connection between the steering wheel and the steering rack of the power steering assembly. Instead, an assembly associated with the steering wheel sends an electronic signal to the power steering assembly. The electronic signal actuates the power steering assembly. Since steer-by-wire systems have no mechanical connection, routing of the rigid shafts between the steering wheel and a steering rack is avoided.

(4) Some known steer-by-wire systems include steering feel motors that utilize torque control to produce a desired steering feel for the vehicle operator. The torque control methodology is intuitive in that the motor will produce a resistance torque feel for the vehicle operator. A low resistance torque (e.g., 6 Nm or less) is typically sufficient for normal driving operations. Though,

occasionally, a high resistance torque (e.g., 15 Nm or higher) may be desirable for parking and/or emergency maneuvers. Therefore, with torque control, a relatively high-power output (e.g., 15-30 Nm) motor is typically provided to produce a desired steering feel for both low-resistant and high-resistant torque events. Because the high-power output motor only provides resistance to the rotation of the steering wheel, the motor does not do positive work and, thus, the energy provided for actuating the motor is converted to heat, noise, vibration, and/or hardness in the steer-by-wire system.

SUMMARY OF THE INVENTION

(5) According to an aspect of the invention, alone or in combination with any other aspect, a steering apparatus is provided for turning steerable wheels of a vehicle. The steering apparatus includes a steering wheel rotatable about a steering axis to effect turning of the steerable vehicle wheels. A first shaft is connected to the steering wheel and rotatable about a first shaft axis. A first nut is threadably connected to and longitudinally movable along the first shaft. A second nut is connected to the first nut such that the first and second nuts are prevented from moving longitudinally relative to one another. A second shaft is rotatable about a second shaft axis. The second nut is threadably connected to and longitudinally movable along the second shaft. A motor is connected to the second shaft and controllable to apply a force to the steering wheel through the first and second shafts and the first and second nuts.

(6) According to an aspect of the invention, alone or in combination with any other aspect, a steering apparatus is provided for turning steerable wheels of a vehicle. The steering apparatus comprises a steering wheel rotatable about a steering axis to effect turning of the steerable vehicle wheels. A first shaft is connected to the steering wheel and rotatable about a first shaft axis. A first nut is threadably connected to and longitudinally movable along the first shaft. A second nut is connected to the first nut via a connector. A second shaft is rotatable about a second shaft axis. The second nut is threadably connected to and longitudinally movable along the second shaft. A coefficient of friction between the second shaft and the second nut is greater than a coefficient of friction between the first shaft and first nut. A motor is connected to the second shaft and controllable to apply a force to the steering wheel through the first and second shafts and the first and second nuts.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other features of the invention will become apparent to one skilled in the art to which the invention relates upon consideration of the following description of the invention with reference to the accompanying drawings, in which:

(2) FIG. **1** is a schematic illustration of a steering apparatus for a motor vehicle;

(3) FIG. **2** is a side view of a portion of the steering apparatus of FIG. **1**; and

(4) FIG. **3** is a schematic illustration of a portion of an example power steering system of the steering apparatus of FIG. **1**.

DETAILED DESCRIPTION OF THE INVENTION

(5) FIG. **1** schematically illustrates a steering apparatus **10** for a motor vehicle. The steering apparatus **10** illustrated in FIG. **1** is a steer-by-wire steering apparatus. However, the teachings of the present disclosure may be adapted for use in non-steer-by-wire steering apparatus. As shown in FIGS. **1-2**, the steering apparatus **10** includes a steering wheel **12**. The steering wheel **12** is of known construction and is manually rotatable by a vehicle operator about a steering axis **14**.

(6) A first bar end **16** of a torsion bar **18** is fixed for rotation with the steering wheel **12**. The first bar end **16** may be directly fixed to the steering wheel **12**, or may be indirectly fixed to the steering wheel **12** via one or more intervening components. A second bar end **20** is fixed for rotation with a

first end **22** of a first shaft **24**. The torsion bar **18** may be formed separately from the first shaft **24** and subsequently connected, or may be formed integrally with the first shaft as one piece. Further, the torsion bar **18** and the first shaft **24** may be directly connected to one another, or indirectly connected to one another by one or more intermediate components.

(7) The first end **22** is supported in a housing **26** via a first mount plate **28** positioned in the housing at a first housing end **30**. A second end **32** of the first shaft **24** is supported in the housing **26** via a second mount plate **34** positioned in the housing between the first housing end **30** and an opposite second housing end **36**. The first shaft **24** is rotatable about a first shaft axis **38** relative to the housing **26** and the mount plates **28**, **34** via bearings **40**, **42** (e.g., a ball bearings) that are positioned between the mount plates and the first shaft. As shown in FIGS. **1-2**, the steering and first shaft axes **14**, **38** may be coaxial. However, the steering and first shaft axes **14**, **38** may be parallel to one another or at a non-zero degree angle relative to one another. The first shaft's connection to the housing **26** via the mount plates **28**, **34** and bearings **40**, **42** is designed such that the first shaft **24** is prevented from moving longitudinally (i.e., in a longitudinal direction along the first shaft axis **38**) during use.

(8) A first nut **44** is threadably connected to threading **46** on the first shaft **24**. The first nut **44** is also connected to a second nut **48** via a plurality of posts **50**. However, the steering apparatus **10** may include any number of posts **50** and/or any other connecting member(s) for connecting the first and second nuts **44**, **48** together.

(9) The posts **50** retain the first and second nuts **44**, **48** at a predetermined distance from one another and, thus, prevent the first and second nuts from moving longitudinally relative to one another. However, because of the posts **50**, longitudinal movement of one of the nuts **44**, **48** in one direction causes the other nut to move in the same direction. The posts **50** not only extend through the second mount plate **34**, but are prevented from rotating about a longitudinal axis (e.g., the first shaft axis **38** and/or a second shaft axis **39**) by the second mount plate, which in turn prevents each of the nuts **44**, **48** from rotating about a longitudinal axis (e.g., the first shaft axis **38** and/or the second shaft axis **39**).

(10) The second nut **48** is threadably connected to threading **52** on a second shaft **54** and movable along the second shaft axis **39** via rotation of the second shaft **54**. As shown in FIGS. **1-2**, the first and second shaft axes **38**, **39** may be coaxial. However, the first and second shaft axes **38**, **39** may be parallel to one another or at a non-zero degree angle relative to one another. A first end **56** of the second shaft **54** is supported in the housing **26** via the second mount plate **34**. A second end **58** of the second shaft **54** is supported in the housing **26** via a third mount plate **60** positioned in the housing at the second housing end **36**. Each of the mount plates **28**, **34**, **50** is fixed to the housing **26** via at least one fastening element **62** (e.g., a screw). The second shaft **54** is rotatable about the second shaft axis **39** relative to the housing **26** and the mount plates **34**, **60** via bearings **64**, **66** (e.g., a ball bearings) that are positioned between the mount plates and the second shaft. The second shaft's connection to the housing **26** via the mount plates **34**, **60** and bearings **64**, **66** is designed such that the second shaft **54** is prevented from moving longitudinally during use.

(11) The second end **58** of the second shaft **54** is fixed for rotation with a drive shaft **68** of a first electric motor **70**. The second end **58** may be formed separately from the drive shaft **68** and subsequently attached thereto, such as, e.g., via the drive shaft being press-fit into the second end. Alternatively, the second end **58** may be formed integrally with the drive shaft **68** as one piece. As another alternative, the second end **58** may be indirectly connected to the first electric motor **70** (e.g., to the drive shaft **68**) via one or more intermediate components. For example, the second end **58** may be indirectly connected to the first electric motor **70** via a gear reduction system. The gear reduction system may include at least a worm wheel and a worm shaft. The first electric motor **70** may be of any known type suitable for use in the steering apparatus **10**. For example, the first electric motor **70** may be a variable reluctance motor, a permanent magnet alternating current motor or a brushless direct current motor.

(12) As shown in FIG. **1**, the first and second shafts **24**, **54** are not directly connected to one another, and may be spaced from one another. Rotation of one of the shafts **24**, **54** thus does not directly cause the other shaft to rotate. However, as will be discussed in more detail below, the shafts **24**, **54** are still operably connected to one another via the nuts **44**, **48** and the posts **50**. Further, the shaft's connection to one another may be such that rotation of one of the shafts **24**, **54** in one direction indirectly causes the other shaft to rotate in the same direction. Alternatively, the threading on the shafts **24**, **54** and/or the nuts **44**, **48** may be configured such that the rotation of one of the shafts in one direction indirectly causes the other shaft to rotate in a second (e.g., opposite) direction. In either case, when one of the shafts **24**, **54** rotates, the other shaft also rotates (however, there may be a relatively minor delay between the start of rotation of one of the shafts and the start of rotation of the other of the shafts).
(13) A steering torque applied by the vehicle operator to the steering wheel **12** urges the first shaft **24** to rotate about the axis **38**, which in turn urges the nuts **44**, **48** to move longitudinally and the second shaft **54** to rotate. The first shaft **24**, however, may be prevented from being rotated by the applied steering torque entirely via friction in the steering apparatus **10**. In particular, a coefficient of friction between the second nut **48** and the second shaft **54** may be selected so that the second nut is prevented from moving longitudinally (and, accordingly, the second shaft is prevented from rotating) when urged to do so by the applied steering torque.
(14) The first nut **44**, through its connection to the second nut **48**, is also prevented from moving longitudinally, which correspondingly prevents the first shaft **24** from rotating. In addition to the friction between the second nut and shaft **48**, **54**, a coefficient of friction between the first nut and shaft **44**, **24** may be selected so that the first nut is at least partially prevented from moving longitudinally (and, accordingly, the first shaft is partially prevented from rotating) when urged to do so by the applied steering torque. Because the first shaft **24** is fixed for rotation with the second bar end **20**, the first shaft prevents at least the second bar end from being rotated about the axis **38** by the steering wheel **12**, which correspondingly provides a predetermined rotational resistance to the steering wheel via the torsion bar **18**.
(15) However, by including the torsion bar **18**, the steering apparatus **10** is configured so that the vehicle operator can rotate the steering wheel **12** about the steering axis **14** relative to the first shaft **24**. For example, the vehicle operator may be able to rotate the steering wheel **12** about 3-6° relative to the first shaft **24**. Therefore, the first bar end **16** thus may twist relative to the second bar end **20** in response to a predetermined applied steering torque, thereby permitting relative rotation between the steering wheel **12** and the first shaft **24**.
(16) The first electric motor **70** may be operated to control the resistance to the rotation of the steering wheel **12** and, accordingly, is commonly referred to as a “steering feel motor.” In particular, actuation of the first electric motor **70** causes the drive shaft **68** to rotate the second shaft **54** about the axis **39**, thus overcoming the friction between the second shaft and the second nut **48**. The second nut **48** moves along the axis **39** in accordance with the direction the second shaft **54** is rotating. The first nut **44**, being connected to the second nut **48**, moves in the same direction as the second nut. The longitudinal movement of the first nut **44** applies a torque (i.e., a rotational force) to the first shaft **24** that may urge the first shaft to rotate about the axis **38**. The first shaft **24**, when rotating, applies a torque to the second bar end **20** (and correspondingly, the entirety of the torsion bar **18**) to provide steering feel. The torque applied by the first shaft **24** may be in the same direction that the vehicle operator rotates the steering wheel **12** about the steering axis **14** and, thus, may make it easier for the vehicle operator to turn the steering wheel. In doing so, the first electric motor **70** overcomes the resistance provided by the friction and reduces the amount of steering torque required by the vehicle operator to turn the steering wheel **12** about the steering axis **14** to a desired position.
(17) The first shaft **24** and the second bar end **20** may thus be rotated by the first electric motor **70** in the same direction as the steering wheel **12** and first bar end **16** about the steering axis **14**.

However, the steering wheel **12** and first bar end **16** may be rotated ahead of the second bar end **20** and the first shaft **24** based on the amount of steering torque that the vehicle operator applies to the steering wheel.

(18) Instead of the rotational resistance to the steering wheel **12** being provided entirely via friction, the first electric motor **70** may also apply rotational resistance to the steering wheel **12**. In such case, the friction between the second nut **48** and the second shaft **54** (and, optionally, the friction between the first nut **44** and the first shaft **24**) partially resists rotation of the first shaft by the applied steering torque in a similar manner as described above. However, because the friction is configured to provide only a portion of the resistance, the applied steering torque may at least partially overcome the friction between the first shaft and nut **24**, **44** and/or the second shaft and nut **54**, **48**. Additional resistance, when desired, may be provided by the first electric motor **70**. In particular, the first electric motor **70** may cause the first shaft **24** to apply a torque to the second bar end **20** (and correspondingly, the entirety of the torsion bar **18**) that is in a direction opposite to that which the vehicle operators steering wheel **12** about the steering axis **14** and, thus, may make it harder for the vehicle operator to turn the steering wheel. In doing so, the first electric motor **70** adds to the frictional resistance to increase the amount of steering torque required by the vehicle operator to turn the steering wheel **12** about the steering axis **14** to a desired position.

(19) The desired coefficient of friction between the first nut and shaft **44**, **24** and/or between the second nut and shaft **48**, **54** may be achieved via material selection, lead angle selection for the threads of the shaft(s), and/or via lead angle selection for inner threads of the nut(s). The threading **48** of the first shaft **24** may have a larger lead angle than the threading **52** of the second shaft **54**. In such a configuration, the coefficient of friction between the second nut and shaft **48**, **54** is greater than the coefficient of friction between the first nut and shaft **44**, **24**. The friction between the second nut and shaft **48**, **54** thus provides more rotational resistance than the friction between the first nut and shaft **44**, **24**.

(20) Having a lower coefficient of friction between the first shaft and nut **44**, **24** may help to reduce the size of the first electric motor **70**. This is because the first electric motor **70** may be controlled to provide “active return” or “return-to-center” functionality for the steering wheel **12**. In order to provide such functionality, the first electric motor **70** has to be powerful enough to drive the second nut and shaft **48**, **54** and the first nut and shaft **44**, **24** to rotate the steering wheel **12** (at a certain speed and at certain acceleration) back to a central alignment position after the steering wheel is rotated. If the coefficient of friction between the first shaft and nut **44**, **24** was as high as the coefficient of friction between the second shaft and nut **48**, **54**, the first electric motor **70** would have to be larger and more powerful in order to provide the “active return” functionality. Therefore, a steering apparatus **10** in which the coefficient of friction between the first shaft and nut **44**, **24** is at least equal to the coefficient of friction between the second shaft and nut **48**, **54** may require a larger and more powerful first electric motor **70** than a steering system apparatus **10** in which the coefficient of friction between the first shaft and nut is less than the coefficient of friction between the second shaft and nut.

(21) As shown in FIG. **1**, a first electronic control unit (“ECU”) **72** is operably connected to the first electric motor **70**. The first ECU **72** actuates and controls the first electric motor **70** to control the steering resistance (i.e., the resistance to the rotation of the steering wheel **12**) provided to the steering wheel. The first ECU **72** is preferably a microcomputer. Alternatively, the first ECU **72** may be formed from discrete circuitry, an application-specific-integrated-circuit (“ASIC”), or any other type of control circuitry.

(22) The steering apparatus **10** may include a torque/position sensor assembly that includes both the torque bar **18** and at least one torque/position sensor **74**. The torque/position sensor assembly may also include a housing in which at least a portion of each of the torque bar **18** and the torque/position sensor **74** are positioned. The torque/position sensor assembly may be structurally and/or operationally positioned between the steering wheel **12** and the first shaft **24**.

(23) The torque/position sensor **74** is operatively connected to the torsion bar **18**, such as, for example, to both the first and second bar ends **16**, **20**. Alternatively or additionally, the torque/position sensor **74** may be operatively connected to a component that is fixed for rotation with the first bar end **16** (e.g., the steering wheel **12**), and/or to a component that is fixed for rotation with the second bar end **20** (e.g., the first shaft **24**). The torque/position sensor **74** may be operable to sense the rotational position (i.e., rotational angle) of the first shaft **24** via the second bar end **20**, the rotational position of the steering wheel **12** via the first bar end **16**, and a relative rotational position (i.e., rotational angle) between the steering wheel **12** and the first shaft via a relative rotational position between the first and second bar ends. The torque/position sensor **74** may be any known sensor or group of sensors for sensing at least the above sensed parameters, and for generating signals indicative of the sensed parameters. The torque/position sensor **74** may include, for example, an optical, a magnetic and/or a mechanical sensor of known construction.
(24) The torque/position sensor **74** may be configured to internally determine a steering torque applied to the steering wheel **12** by the vehicle operator as a function of a spring constant of the torsion bar **18** and the relative angle between the first and second bar ends **16**, **20**. In such case, the signals generated by the torque/position sensor **74** would include the vehicle operator applied steering torque.
(25) A second ECU **76** is operatively coupled to the torque/position sensor **74** and to the first ECU **72**. The second ECU **76** receives the signals indicative of the rotational position of the first shaft **24** from the torque/position sensor **74** and the relative rotational position between the steering wheel **12** and the first shaft. The second ECU **76** may also receive the signals from the torque/position sensor **74** indicative of the vehicle operator applied steering torque when determined by the torque/position sensor. Alternatively, the second ECU **76** may receive signals indicative of the relative angle between the steering wheel **12** and first shaft **24** and internally determine the applied steering torque as a function of those received signals and the spring constant of the torsion bar **18**. In response to the signals from the torque/position sensor **74**, the second ECU **76** generates and transmits a first signal corresponding to the sensed vehicle operator applied steering torque. The second ECU **76** may also be operatively connected to and communicate with a vehicle CAN bus **78**. The second ECU **76** may request and/or receive certain vehicle operating characteristics, such as vehicle speed, from the vehicle CAN bus **78**. The second ECU **76** is preferably a microcomputer. Alternatively, the second ECU **76** may be formed from discrete circuitry, an application-specific-integrated-circuit (“ASIC”), or any other type of control circuitry.
(26) The steering apparatus **10** also includes a power steering system **80** configured to turn steerable vehicle wheels **82**. In the example configuration of FIGS. **1** and **3**, the power steering system **80** is an electric power steering system. However, the steering apparatus **10** can be configured to include other types of power steering systems besides electric (e.g., hydraulic).
(27) The power steering system **80** includes a third ECU **84**. The third ECU **84** receives the first signal from the second ECU **76**. The third ECU **84** is further operatively coupled to a second electric motor **86**. The third ECU **84** controls the operation of the second electric motor **86** in accordance with the first signal. For example, the third ECU **84** may run an algorithm that uses the vehicle operator applied steering torque from the first signal as a parameter for controlling the operation of the second electric motor **86**. The third ECU **84** is preferably a microcomputer. Alternatively, the third ECU **84** may be formed from discrete circuitry, an application-specific-integrated-circuit (“ASIC”), or any other type of control circuitry. Although the power steering system **80** is shown and described as having a single electric motor **86** controlled by a single ECU **84**, the power steering system may have any number of electric motors and ECUs for turning the steerable vehicle wheels **82**.
(28) The second electric motor **86** may be of any known type suitable for use in the steering apparatus **10**. For example, the second electric motor **86** may be a variable reluctance motor, a permanent magnet alternating current motor or a brushless direct current motor.

(29) The second electric motor **86** effects turning of the steerable vehicle wheels **82** through a gear assembly **88**. FIG. **3** schematically depicts an example gear assembly **88** that is operably connected to the second electric motor **86**. The gear assembly **88** of FIG. **3** includes a worm screw **90** that is fixed for rotation with a drive shaft **92** of the second electric motor **86**. The worm screw **90** may be formed separately from the drive shaft **92** and subsequently attached, or may be formed integrally with the drive shaft as one piece. The drive shaft **92** extends along a drive shaft axis **94**. The worm screw **90** rotates with the drive shaft **92** about the drive shaft axis **94**.

(30) A worm wheel **96** meshingly engages and is rotatable by the worm screw **90**. The worm wheel **96** is fixed for rotation with a first pinion end **98** of a pinion **100**. The worm wheel **96** may be formed separately from the first pinion end **98** and subsequently attached, or may be formed integrally with the first pinion end. The pinion **100** extends along a pinion axis **102**. The first pinion end **98** rotates with the worm wheel **96** about the pinion axis **102**.

(31) A second pinion end **104** has a gear portion **106** that is meshingly engaged to a gear portion **108** of a steering rack **110**. Rotation of the pinion **100** about the pinion axis **102** causes the steering rack **110** to move linearly along a rack axis **112** relative to the vehicle. As shown in FIG. **1**, each end of the steering rack **110** is connected to an associated steerable wheel by a tie rod **114**. Therefore, the linear movement of the steering rack **110** acts on the steerable vehicle wheels **82** through the tie rods **114** to turn the steerable vehicle wheels. The steering rack **110** may be supported by a housing **116** and may linearly move relative to the housing. At least one of the components of the gear assembly **88** may be supported in or on the housing **116**.

(32) The second electric motor **86** thus effects the turning of the steerable vehicle wheels **82** in accordance with to the first signal through the gear assembly **88**. The number and type of components in the gear assembly **88** described and shown in the present disclosure is merely just one example gear assembly used to operably connect the second electric motor **86** to the steering rack **110**. The gear assembly **88** may include any number and type of components necessary for operably connecting the second electric motor **86** to the steering rack **110**.

(33) As shown in FIGS. **1** and **3**, the power steering system **80** may include at least one position sensor **118** for sensing a linear position of the steering rack **110** along the rack axis **112**. The position sensor **118** may be any known sensor or group of sensors for sensing a linear position of the steering rack **110** along the rack axis **112** and for generating signals indicative of the sensed linear position of the steering rack. The position sensor **118** may include, for example, an optical, a magnetic and/or a mechanical sensor of known construction. As shown in FIG. **1**, the position sensor **118** is operatively connected to the second ECU **76**. The second ECU **76** receives the signals indicative of the linear position of the steering rack **110** from the position sensor **118**.

(34) In the illustrated embodiment, the torque/position sensor **74** is electrically connected to the third ECU **84** by the second ECU **76**. The second ECU **76** generates and transfers the first signal to the third ECU **84**. The third ECU **84** controls a torque output of the second electric motor **86** in accordance with the vehicle operator applied steering torque provided in the first signal. For example, the third ECU **84** may control the amount of torque outputted by the second electric motor **86** and the direction of the outputted torque. The second electric motor **86** thus is torque controlled (i.e., controlled as a function of the vehicle operator applied steering torque).

(35) The second ECU **76** receives the signals indicative of the linear position of the steering rack **110** from the position sensor **118**. Using the linear position of the steering rack **110**, the second ECU **76** determines a rotational position of the first shaft **24** that directly corresponds to the linear position of the steering rack. The determination may be made by running algorithms that transform the received linear position of the steering rack **110** into a corresponding rotational position of the first shaft **24**. Alternatively, the determination may be made by accessing stored reference data that includes predetermined linear positions of the steering rack **110** and predetermined rotational positions of the first shaft **24** that directly correlate to the predetermined linear positions of the steering rack.

(36) The second ECU **76** then compares a current rotational position of the first shaft **24** (i.e., the rotational position of the first shaft that was sensed by the torque/position sensor **74** and received by the second ECU) with the determined rotational position of the first shaft that directly corresponds to the linear position of the steering rack **110**. The second ECU **76** generates and transmits a second signal corresponding to the difference between the current rotational position of the first shaft **24** and the determined rotational position of the first shaft that directly corresponds to the linear position of the steering rack **110**. The second signal is transmitted to the first ECU **72**.
(37) The first ECU **72** receives the second signal. In response to the second signal, the first ECU **72** controls the first electric motor **70** to rotate the first shaft **24** as a function of the difference between the first shaft's current position and the determined rotational position of the first shaft that directly corresponds to the linear position of the steering rack **110** so that the first shaft may be rotated to the rotational position that directly corresponds to the linear position of the steering rack. Therefore, the first electric motor **70** is position controlled (i.e., controlled as a function of the linear position of the steering rack **110**).
(38) The sensors **74**, **118**, the first electric motor **70** and the first and second ECUs **72**, **76** may thus function as a closed loop position control for the first shaft **24** in that the position sensor measures the linear position of the steering rack **110**, the second ECU receives a signal from the position sensor, the second ECU transmits the second signal to the first ECU, the first ECU controls the first electric motor to rotate the first shaft from a first rotational position measured by the torque/position sensor to a rotational position that directly corresponds to the linear position of the steering rack. The first electric motor **70** may be controlled to rotate the first shaft **24** until the torque/position sensor **74** senses that the first shaft is in the rotational position that directly corresponds to the linear position of the steering rack **110**.
(39) The first shaft **24** rotating to the rotational position that directly corresponds to the linear position of the steering rack **110** applies torque to the second bar end **20** (and correspondingly, to the steering wheel **12** through the torsion bar **18**). The first shaft **24** may be rotated in the same direction that the vehicle operator rotates the steering wheel **12** about the steering axis **14** and, thus, may reduce the amount of steering torque required by the vehicle operator to turn the steering wheel about the steering axis to a desired position. The first electric motor **70**, by rotating the first shaft **24** to the position that directly corresponds to the linear position of the steering rack **110**, may thus reduce the amount of steering resistance provided to the steering wheel **12**.
(40) How much the first electric motor **70** reduces the amount of steering resistance may be a function of the speed at which the first shaft **24** is rotated by the first electric motor. The first ECU **72** may be configured to determine and control the operating speed of the first electric motor **70**. Alternatively, the second ECU **76** may be configured to determine the operating speed of the first electric motor **70** and transmit that operating speed to the first ECU **72** for controlling the first electric motor. The determined operating speed of the first electric motor **70** may be a function of the rotational distance that the first shaft **24** needs to rotate in order to assume the rotational position that directly corresponds to the linear position of the steering rack **110**, the speed at which the second electric motor **86** drives the steering rack **110**, the amount of vehicle operator applied steering torque, the type of driving maneuver being performed by the vehicle operator (e.g., parking, turning, and/or slight veering), and/or other vehicle characteristics, such as the vehicle speed.
(41) In case of a failure of the first shaft **24**, the first nut **44**, the second nut **48**, the posts **50**, the second shaft **54**, the first electric motor **70** and/or the first ECU **72**, the first electric motor may not be functional to rotate the first shaft to a rotational position that corresponds to the linear position of the steering rack **110**. In this case, the vehicle operator applied steering torque is still determined and sent to the power steering system **80** for controlling the steerable vehicle wheels **82** accordingly. Operation of the power steering system **80**, the steering apparatus **10** and/or the vehicle may be adjusted in response to the failure.

(42) Instead of being torque controlled, the second electric motor **86** may be positioned controlled. In such case, the torque/position sensor **74** may transmit a signal indicative of the rotational position of the steering wheel **12** to the second ECU **76**, and the position sensor **118** may transmit a signal indicative of the linear position of the steering rack **110** to the second ECU. Using the rotational position of the steering wheel **12**, the second ECU **76** determines a linear position of the steering shaft **110** that directly corresponds to the linear position of the steering wheel. The determination may be made by running algorithms that transform the rotational position of the steering wheel **12** into a corresponding linear position of the steering shaft **110**. Alternatively, the determination may be made by accessing stored reference data that includes predetermined rotational positions of the steering wheel **12** and predetermined linear positions of the steering rack **110** that directly correlate to the predetermined rotational positions of the steering wheel.
(43) The second ECU **76** then compares a current linear position of the steering rack **110** (i.e., the linear position of the steering rack that was sensed by the position sensor **118** and received by the second ECU) with the determined linear position of the steering rack that directly corresponds to the rotational position of the steering wheel **12**. The second ECU **76** generates and transmits a signal corresponding to the difference between the current linear position of the steering rack **110** and the determined linear position of the steering rack that directly corresponds to the rotational position of the steering wheel **12**. The signal is transmitted to the third ECU **84**.
(44) The third ECU **84** receives the signal. In response to the signal, the third ECU **84** controls the second electric motor **86** to linearly move the steering rack **110** as a function of the difference between the steering rack's current position and the determined linear position of the steering rack that directly corresponds to the rotational position of the steering wheel **12** so that the steering rack may be moved to the linear position that directly corresponds to the rotational position of the steering wheel. The second electric motor **86** may be controlled to linearly move the steering rack **110** until the position sensor **118** senses that the steering rack is in the linear position that directly corresponds to the rotational position of the steering wheel. Therefore, the second electric motor **86** may be position controlled (i.e., controlled as a function of the rotational position of the steering wheel **12**).
(45) Instead of being position controlled, the first electric motor **70** may be controlled as a function of an estimated steering rack load. In such case, a current of the second electric motor **86** may be sensed internally and/or via the third ECU **84**. The current of the second electric motor **86** may directly or indirectly correspond to how much effort it takes for the second electric motor to linearly move the steering rack **110** to the appropriate linear position. The third ECU **84** transmits a signal indicative of the current to the second ECU **76**. In response to the signal from the third ECU **84**, the second ECU **76** determines an estimated steering rack load as a function of the current of the second electric motor **86**. The determination may be made by running algorithms that transform the current of the second electric motor **86** into a corresponding estimated steering rack load. The second ECU **76** then generates and transmits a signal indicative of the estimated steering rack load to the first ECU **72**.
(46) The first ECU **72** receives the signal and controls a torque output of the first electric motor **70** in accordance with the estimated steering rack load provided in the signal. For example, the first ECU **72** may run an algorithm that uses the estimated steering rack load from the signal as a parameter for controlling the operation of the first electric motor **70**, and then control the amount and the direction of torque outputted by the first electric motor accordingly. The first electric motor **70** thus may be steering rack load controlled (i.e., controlled as a function of the estimated steering rack load). Further, because the torque output of the first electric motor **70** corresponds to the estimated steering rack load, the torque applied to the steering wheel **12** via the first electric motor also corresponds to the estimated steering rack load. The torque output of the first electric motor **70** may be monitored via the torque/position sensor **74** (and adjusted in accordance with the sensed measurements of the torque/position sensor **74**) to help produce a correspondence between the

estimated steering rack load and the torque applied to the steering wheel **12** via the first electric motor **70**.

(47) In addition to providing steering feel, the first electric motor **70** can be controlled via the first ECU **72** to limit the degree to which the steering wheel **12** may be rotated in each direction about the steering axis **14** and, thus, may function in addition to or in place of a separate mechanical rotational end stop for the steering wheel. In particular, once the steering wheel **12** is rotated a predetermined degree in one direction, the first ECU **72** may control the first electric motor **70** to prevent any further rotation of the steering wheel **12** in the same direction. In one example configuration, the first ECU **72** may "turn off" the first electric motor **70** so that the friction between at least the second nut and shaft **48**, **54** prevents any further rotation of the steering wheel **12** in the one direction. Alternatively, the first ECU **72** may drive the first electric motor **70** to provide a torque to the steering wheel **12** that acts in concert with the frictional resistance to prevent any further rotation of the steering wheel in the one direction.

(48) From the above description of the invention, those skilled in the art will perceive improvements, changes and modifications. For example, although the power steering system **46** is described and shown as having a rack-and-pinion drive, the power steering system may include any other type of drive, such as, for example, a belt drive and/or a drive that utilizes a ball-nut assembly.

(49) Algorithms, determinations, and/or computing processes for controlling the first electric motor **70** may be performed in either the first ECU **72** and/or the second ECU **76**. Algorithms, determinations, and/or computing processes for controlling the second steering electric motor **86** may be performed in either the third ECU **84** and/or the second ECU **76**.

(50) The steering apparatus **10** may be configured with only the first and third ECUs **72**, **84**, and without the second ECU **76**. The first and third ECUs **72**, **84** would thus perform all of their respective algorithms, determinations, computing processes, control functions and directly communicate with one another. In this case, the first ECU **72** may be operatively connected to the torque/position sensor **74**, generate the first signal, and transmit the first signal. Similarly, the third ECU **84** may be operatively connected to the position sensor **118**. The third ECU **84** may generate a signal indicative of the linear position of the steering rack **110**, and transmit that signal to the first ECU **72**. The first ECU **72**, after receiving the signal from the third ECU **84**, may then perform all processes necessary for controlling the first electric motor **70**.

(51) The functions of the ECUs **72**, **76**, **84** can be divided between the ECUs as described above, or can be distributed among the ECUs in a different manner. The three ECUs **72**, **76**, **84** can be located at any physical location in the vehicle. The three ECUs **72**, **76**, **84** can be combined into two or one ECU located at any physical location in the vehicle.

(52) Also, although a torsion bar **18** has been provided to permit the steering wheel **12** to rotate relative to the second gear, the steering apparatus **10** may utilize structures other than the torsion bar **18** to permit such relative movement. Such improvements, changes and modifications within the skill of the art are intended to be covered by the appended claims.

Claims

1. A steering apparatus for turning steerable wheels of a vehicle, the steering apparatus comprising: a steering wheel rotatable about a steering axis to effect turning of the steerable vehicle wheels; a first shaft connected to the steering wheel and rotatable about a first shaft axis; a first nut threadably connected to and longitudinally movable along the first shaft; a second nut connected to the first nut such that the first and second nuts are prevented from moving longitudinally relative to one another; a second shaft rotatable about a second shaft axis, the second nut being threadably connected to and longitudinally movable along the second shaft; and a motor connected to the second shaft and controllable to apply a force to the steering wheel through the first and second

shafts and the first and second nuts.
2. The steering apparatus recited in claim 1, wherein rotation of the second shaft via the motor causes the second nut to move along the second shaft in a first direction, movement of the second nut in the first direction causing the first nut to move along the first shaft in the first direction.
3. The steering apparatus recited in claim 2, wherein the first nut moving along the first shaft applies a torque to the first shaft that urges the first shaft to rotate about the first shaft axis.
4. The steering apparatus recited in claim 1, a lead angle of threading of the first shaft being greater than a lead angle of threading of the second shaft.
5. The steering apparatus recited in claim 1, wherein a coefficient of friction between the first nut and the first shaft being less than a coefficient of friction between the second nut and the second shaft.
6. The steering apparatus recited in claim 1, wherein the first and second nuts are connected and prevented from moving relative to one another via a post.
7. The steering apparatus recited in claim 1, wherein the first and second shafts are spaced from one another.
8. The steering apparatus recited in claim 1, wherein the motor is controllable to limit the degree to which the steering wheel may be rotated in each direction about the steering axis.
9. The steering apparatus recited in claim 1, further comprising a housing, the first nut, the second nut, a portion of the first shaft and a portion of the second shaft being in the housing, the first and second shafts being rotated about the first and second shaft axes relative to the housing.
10. The steering apparatus recited in claim 1, wherein rotational resistance to the steering wheel is provided entirely via friction.
11. The steering apparatus recited in claim 1, wherein rotational resistance to the steering wheel is provided via both friction and the motor.
12. The steering apparatus recited in claim 1, wherein the first and second shaft axes are coaxial.
13. The steering apparatus recited in claim 1, further comprising a torsion bar having a first bar end connected to the steering wheel and a second bar end connected to the first shaft, the first bar end twisting relative to the second bar end in response to a predetermined steering torque applied to the steering wheel, the twisting of the torsion bar permitting the steering wheel to rotate about the steering axis relative to the first shaft.
14. The steering apparatus recited in claim 1, wherein the motor controls the amount of steering resistance provided to the steering wheel by applying a force to the steering wheel through the first and second shafts and the first and second nuts.
15. The steering apparatus recited in claim 1, further comprising: a first sensor for determining a steering torque applied to the steering wheel by the vehicle operator; and a power steering system configured to turn the steerable vehicle wheels in accordance with the determined steering torque, the power steering system having a second sensor for sensing a position of a portion of the power steering system, the motor being controlled in accordance with the sensed position.
16. The steering apparatus recited in claim 15, wherein the power steering system includes a steering rack connected to the steerable vehicle wheels and linearly movable along a rack axis relative to the vehicle, the second sensor sensing a linear position of the steering rack along the rack axis, the motor being controlled in accordance with the sensed linear position of the steering rack.
17. The steering apparatus recited in claim 15, wherein the power steering system includes a second motor operably connected to a steering rack and configured to linearly move the steering rack along a rack axis, an output of the second motor being controlled in accordance with the determined steering torque.
18. The steering apparatus recited in claim 1, further comprising: a first sensor for determining a rotational position of the steering wheel; and a power steering system configured to turn the steerable vehicle wheels in accordance with the determined rotational position of the steering

wheel, the power steering system including a second motor operably connected to a steering rack and configured to linearly move the steering rack along a rack axis, an estimated steering rack load of the steering rack being determined as a function of a current of the second motor, the motor being controlled in accordance with the estimated steering rack load.

19. The steering apparatus recited in claim 1, wherein the steering apparatus is a steer-by-wire steering apparatus.

20. A steering apparatus for turning steerable wheels of a vehicle, the steering apparatus comprising: a steering wheel rotatable about a steering axis to effect turning of the steerable vehicle wheels; a first shaft connected to the steering wheel and rotatable about a first shaft axis; a first nut threadably connected to and longitudinally movable along the first shaft; a second nut connected to the first nut via a connector; a second shaft rotatable about a second shaft axis, the second nut being threadably connected to and longitudinally movable along the second shaft, a coefficient of friction between the second shaft and the second nut being greater than a coefficient of friction between the first shaft and first nut; and a motor connected to the second shaft and controllable to apply a force to the steering wheel through the first and second shafts and the first and second nuts.